

VersusHQ Update Tournaments & More!

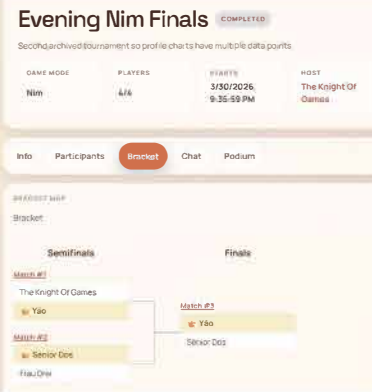


NEW FEATURES

- BRACKET STYLE TOURNAMENTS
- NEW GAME: BATTLESHIP
- PROFILE STATISTICS TRACKING
- NEW & IMPROVED UI

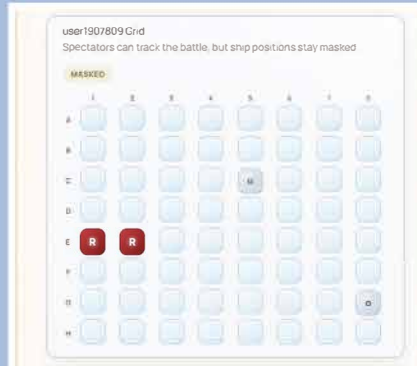
TOURNAMENTS

Test your skills in classic and new VersusHQ games against other players in the new lvl bracket-style tournaments! Accumulate points from victories for bragging rights on the new global leaderboard!



BATTLESHIP

Play against a friend or hone your skills against an ai in our new Battleship game mode! Battleship is also available in the newly added Tournament system!



DESIGN DECISIONS:

The NiteOwls team understands our communities desire for competition, so for our first large update we decided that tournaments with stat tracking and a new gamemode would keep things fresh.

UPCOMING UPDATE:

- Private tournaments & private groups w/ chatrooms
- Friend system to easily challenge and message friends
- New performance stats to track for XP and site-wide levels

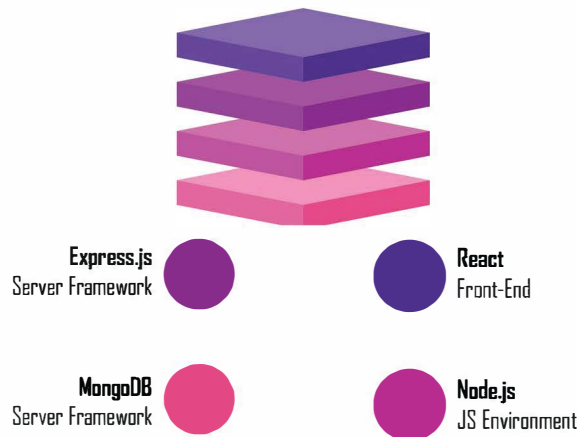
PROFILE ENHANCEMENTS

Log in via Google SSD and visit your and other player's newly customizable profiles to view performance statistics.



TECHNOLOGY STACK

VersusHQ is built using a MERN stack forked off of our parent site, GameNite. This stack structure was selected to easily build off of the source codebase, allowing for simplified database management and a clean API integration.



RESOURCES:

TEAM NiteOwls:

- Caio DaSilva
- Ananya Saggi
- John Rotondo
- Andrea Son



[Click here to try out Versus HQ for yourself](#)



[Click here for our Github repo for the source code](#)



[Click here for a video demo of Versus HQ](#)



PlayPal

Developed by Nour Tawfik, Olivia Romito,
Nadeem Baba, Maya Karintholil

Group 203

**Play Games. Make Friends. Build
Community.**

Account & Profiles

Customize your profile!

Upload a photo or pick an avatar, write a bio, add interest tags, and show off your game stats and win ratios — all in one place. Your profile, your identity.

Friend System

Meet someone cool? Add them!! Send and accept friend requests to build your network of gaming connections. Check your friend list and visit anyone's profile with a click.

Messaging & Chat

Stay connected — in and out of the game!

Real-time direct messages and group chats powered by WebSockets let you strategize with friends, coordinate sessions, and talk whenever you want. Plus, chat with your opponent live during games and vote on polls mid-match!

Game Forums

Every game deserves its own conversation.

Dedicated forum sections per game with category banners, markdown formatting, image uploads, likes/upvotes, and @mentions with friend autocomplete. Find your people, share strategies, start discussions.

Safety & Security

Play safe, stay safe!!

Secure OAuth login with email verification keeps your account protected, while role-based moderation tools let moderators review reports, remove content, and warn or suspend bad actors. See something off? Flag it and we'll handle the rest.

Why these features?

GameNite had great games but no way to actually connect with the people you're playing against. We built the social layer — profiles, messaging, forums, friends — to turn it from a game site into a gaming community where players don't just compete, they belong.

Link to: [PlayPal Website](#)

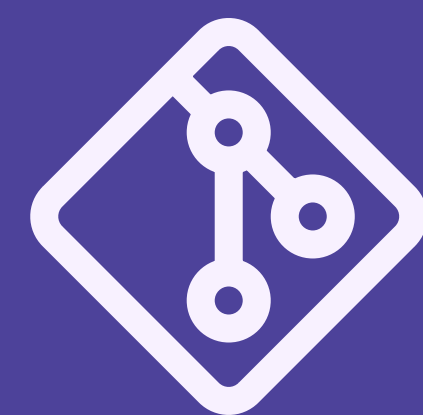
Link to: [Git Repository](#)

CS4530 FINAL PROJECT: GAMENITE!

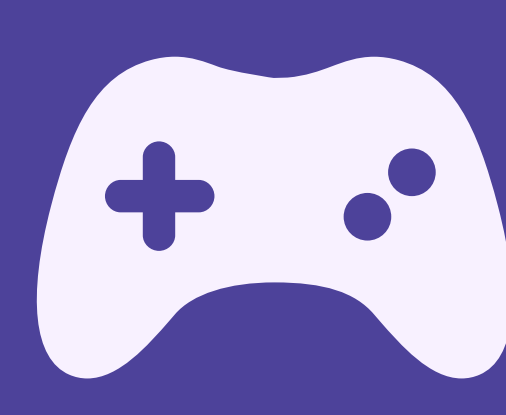
Group 204: Kashvi, Tanisha, Aunnie, Ha

OUR FEATURES

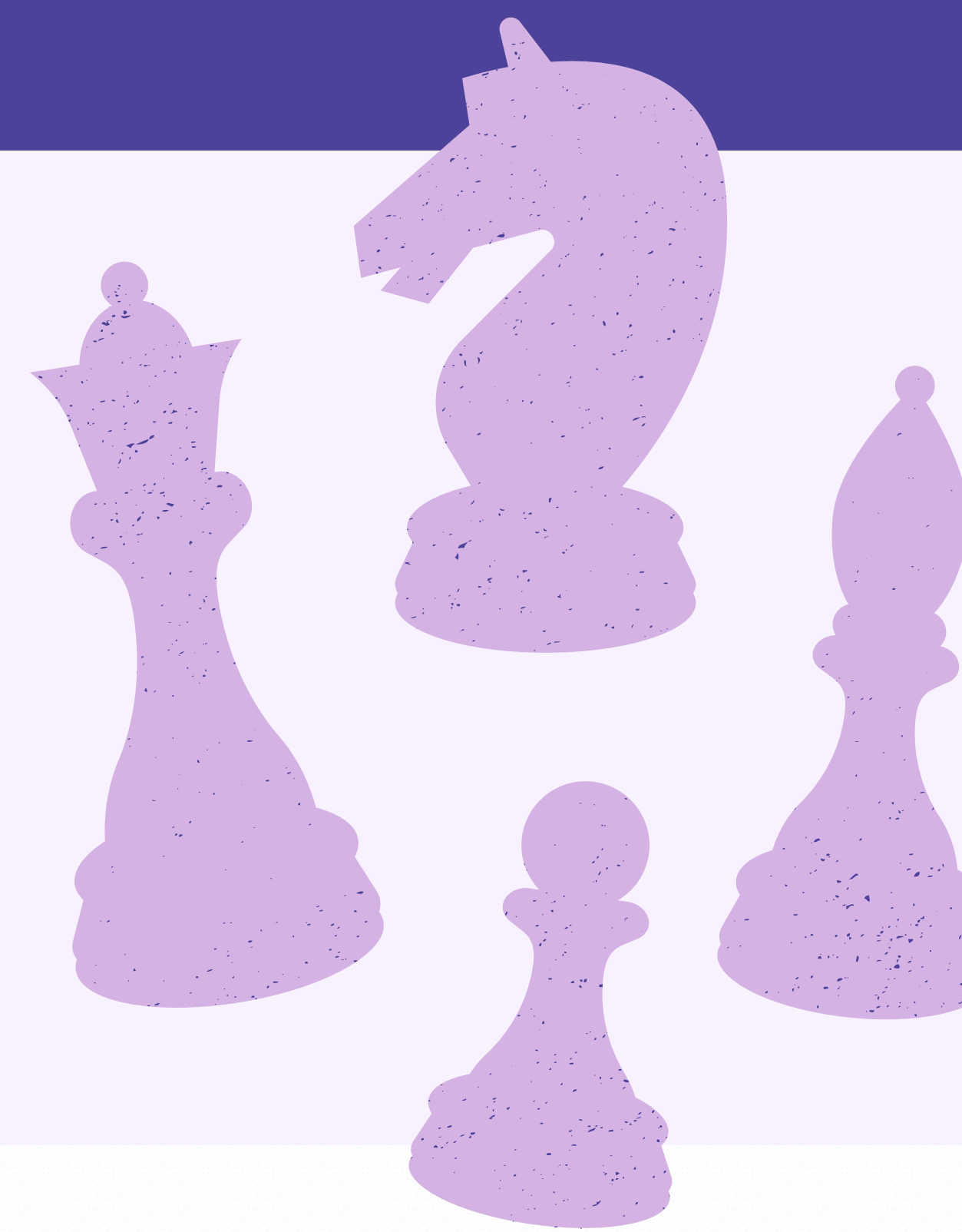
GameNite is a gaming platform that allows users to play games but it lacks the infrastructure needed for competitive and social interaction. Our project enhances GameNite by transforming it into a more engaging, community-driven experience. Inspired by platforms like Chess.com, we introduce features that promote competition and social connectivity. These include real-time leaderboards that allow users to track their rankings globally and among friends, a friend system that enables users to connect and play together, and detailed match history tracking that provides insight into player performance over time. We also implemented a skill-based rating and league system using an Elo-based model, allowing users to compete with others at similar skill levels and track their progression through leagues such as Bronze, Silver, and Gold. Additionally, enhanced user profiles allow players to showcase their achievements, statistics, and activity, helping foster a stronger sense of identity and community within the platform. We also integrated Single Sign-On (SSO), allowing users to securely log in and maintain a consistent identity across sessions. We also implemented a full Chess experience including an AI opponent with three difficulty levels, timed matches with 5, 10, or 30 minute controls, post-game statistics tracking captures and time used, move-by-move game replay accessible from Match History, and the ability to resign at any point during a match.



REPO



DEMO



Chess

Game room created seconds ago

Player #1 is [Frau Drei](#)

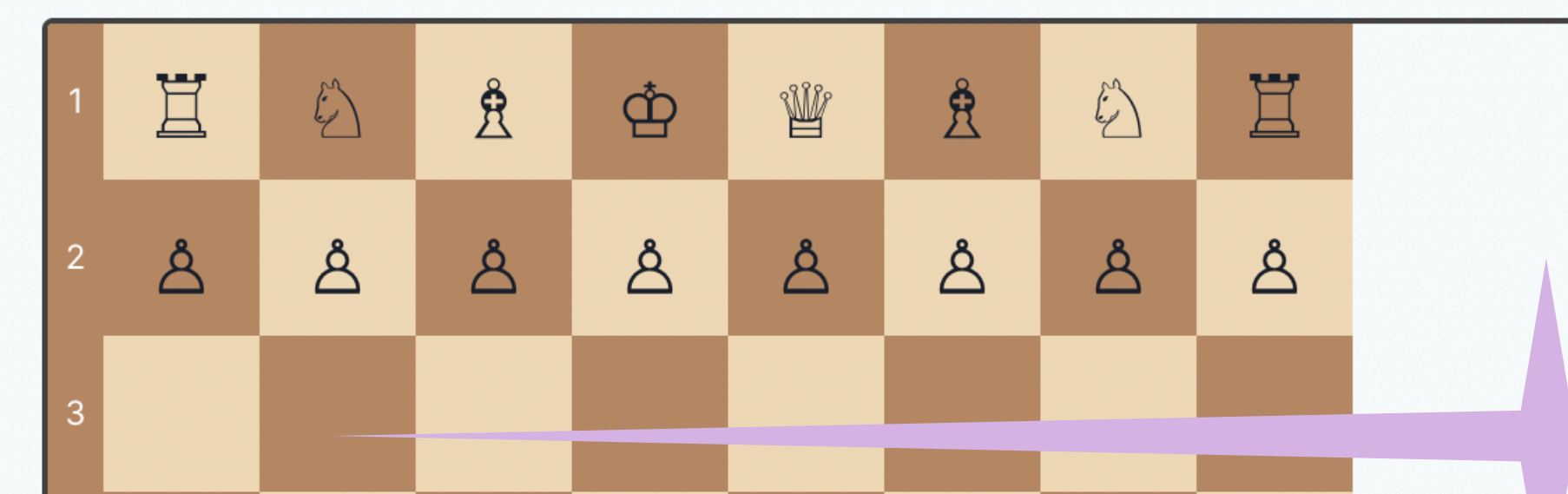
you are player #2

Chess

☐ White: Frau Drei

☒ Black: The Knight Of Games

Waiting for Frau Drei...



Welcome back!

Sign in to continue to GameNite.

user0

.....

☐ Show Password

Log In

or

Continue with Google

Don't have an account? [Create New Account](#)

GameNite stores passwords in plaintext — please don't reuse passwords here.

DEMO

Our demo (<https://spring26-project-group-204-1.onrender.com/login>) showcases the full user experience of the enhanced GameNite platform. Users can play chess against human opponents or an AI at varying difficulty levels, use timed matches with automatic time-loss detection, and review completed games move-by-move in Match History. The demo also highlights the friend system, real-time leaderboard updates, league placements, player profiles, and the ability to compete in both public and private matches.

FUTURE WORK

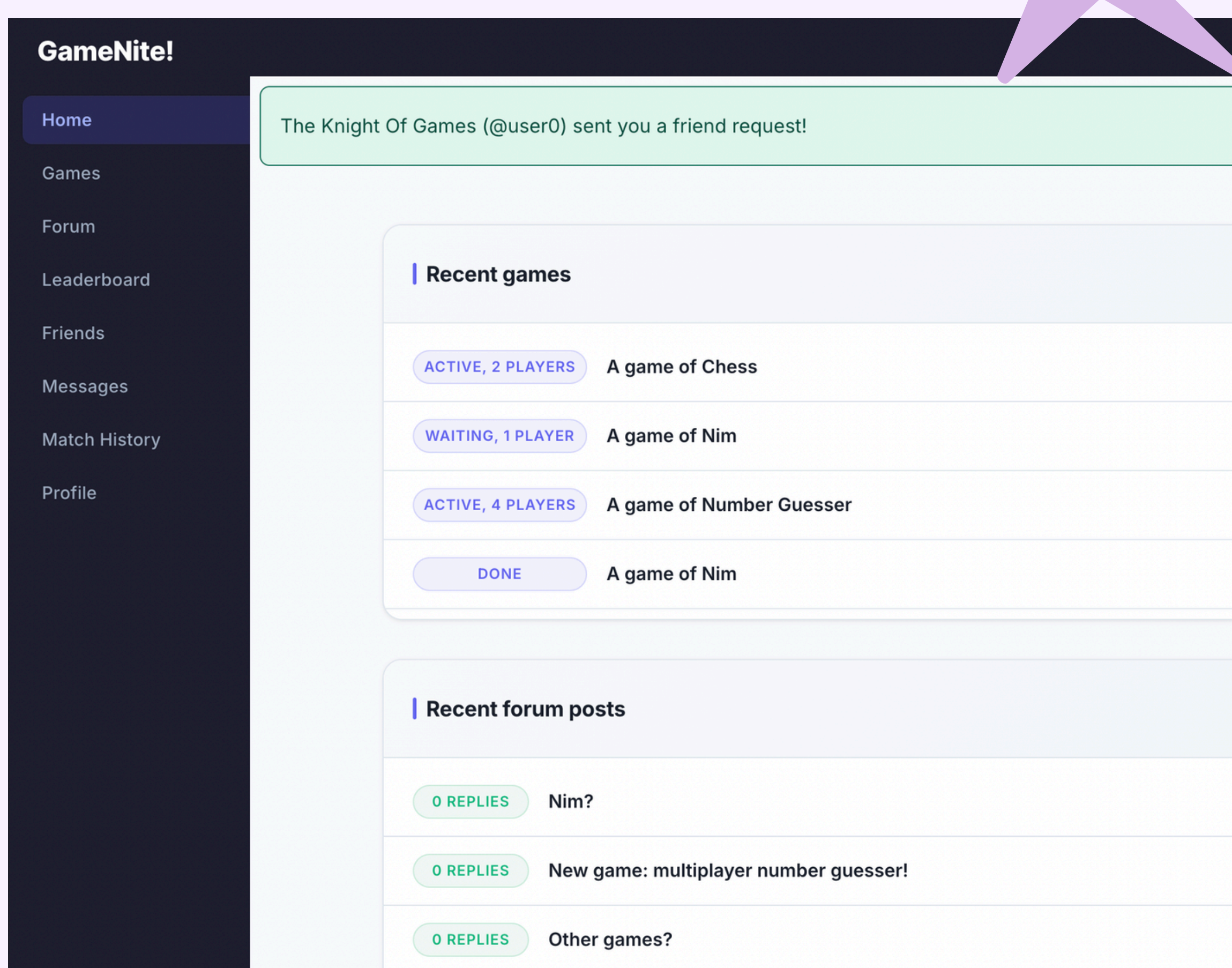
In the future, we aim to expand the platform with additional features to further enhance user engagement and gameplay experience. We plan to introduce additional game types beyond Chess, Nim, and Number Guesser to grow the platform's variety. Additional improvements include a more robust notification system to alert users of friend requests, game invitations, and match results in real time. We would also like to improve mobile responsiveness and expand privacy controls, giving users more flexibility over how they interact with the platform. Further chess enhancements could include opening book analysis, endgame tablebase integration, and tournament-style brackets. These enhancements will help continue evolving GameNite into a fully featured competitive and social gaming environment.

OUR TECHNOLOGY STACK & DESIGN

We implemented our features within the existing GameNite codebase using a modern full-stack architecture. Our frontend is built with React, enabling a dynamic and responsive user interface, while the backend uses Node.js and Express to handle API requests and business logic.

We designed scalable database schemas to support leaderboards, match history, user ratings, and the friend system, ensuring efficient data storage and retrieval. Real-time updates were incorporated into the leaderboard and notification systems to reflect changes immediately after matches are completed. Chess move validation, rule enforcement, and PGN generation are handled using the chess.js library, ensuring accurate and efficient game logic on both the client and server.

In addition, we improved the overall user interface by introducing a more cohesive color scheme and cleaner visual design, making the platform more intuitive and visually appealing. Our system follows a modular design, allowing different components such as profiles, leaderboards, and game logic to interact seamlessly.



Leaderboard

Chess

Nim

Guess

All Players

All Leagues

Bronze

Silver

Gold

Chess Leaderboard

RANK	PLAYER	RATING	LEAGUE
#1	Frau Drei @user3	1016	bronze
#2	You @user0	984	bronze

SOCIALSPHERE

Group 205: Sarah LaCroix, Suhani Kashyap, Vithya Srikumar, Elise Lamson



What's New?

NEW

Welcome to SocialSphere! We've added tons of new features with a focus on socialization:

- More profile customization
- New forum features + trending forum posts
- Group chats + private messages
- Leaderboards
- Admin users

Profiles

Profiles now support more customization options including profile photos, bios, private accounts, two-factor authentication, tracking profile views, and creating daily game and time limits. You can also log in as an admin user to moderate disallowed words, hide threads and mute users.

My Profile

Recent profile views: user0, user2



Yāo

@user1

"your bio goes here!"

Created: 4 hours ago

Make Private

Enable 2FA

Forum

The forum page now shows which forum posts are trending, which is based on its number of upvotes. Users may hide or unhide specific posts. Within the forum posts, there is now support for nested comments, upvoting, editing and deleting. When someone replies to your forum post, it will show up in the inbox.

no way, social sphere is done already?

Reply by Sénior Dos seconds ago ▲ 1 Reply

yep!

Reply by you OP just now ▲ 0 Reply Edit Delete

Leaderboard

#	Username	Wins
1	you	17
2	Bob Martinez	12
3	Jordan Patel	8
4	Sénior Dos	8
5	Alyssa Nguyen	3
6	Morgan Reed	2
7	Taylor Kim	0

Leaderboard

The new Leaderboard page allows you to compare your game stats to other users. You can choose to view the leaderboard for all users or just friends, select different time ranges, select the leaderboard for a specific game, and sort the leaderboard by any of the game stats.

Groupchats

Your Groupchats

Create New Groupchat

Groupchats

• Join testChat

Created 2 minutes ago

Most recent message: ""hello new users!"" - The Kr

Users can now create groupchats, which show up on the new groupchat page with a flag if they are unread. In these chats they can send images and private game invites, and there is more message functionality including reacting to messages and sending emojis. Users can also private message users through their profiles.

Tech Stack + Design

This project was built with a React/TypeScript frontend, utilizing GitHub Actions for CI/CD and deployed via Render. Our features were focused on the socialization of users, which we found were lacking on Gamenite – users couldn't show off their personality in profiles and messages, message outside of games, or fully interact in the forums. This motivated our design for building off of the existing pages and add new pages such as the leaderboard and inbox.

What's Next?

In the future, it would be interesting to add more features to SocialSphere to further the user's capabilities. Some of our extension ideas included being able to view which users are online, seeing when users are typing in groupchats, searching through groupchat messages, favoriting forum posts, and extending admin abilities to be able to ban users.

Links

Check out SocialSphere!

Github

Render



What is GuildSpaces?

GuildSpaces! offers browser-based games, extensive social features, and all of the tools you need to start your own thriving community. Think of it as a centralized space for players to discover others, organize groups, or share gaming content.

The GuildSpaces! experience was designed around three key pillars:

- Community & Moderation
- Social Networking
- Engagement & Content

Pillar 1: Community & Moderation

- Game-specific guilds with banner images, descriptions, member counts
- Rich post creation: titles, body (rich text), tags, spoiler toggle, drafts, polls, media attachments
- Tiered role system: Owner → Moderator → Member with canModerate() utility
- Auto-ban rules: keyword/phrase matching with configurable scope & action
- AI-assisted moderation: in-process Naive Bayes classifier flags toxic content for review
- Pin posts, delete content, ban/unban users from guild pages

Pillar 2: Social Networking

- User search by username + friend request system with notifications
- Real-time 1-to-1 direct messaging with full history (Socket.IO)
- Group chats: create, add/remove friends, leave groups
- Block list management: blocks remove friendship & prevent all contact
- Private game rooms: password-protected, friends bypass password
- Player profiles: XP, 10-tier leveling, medals, per-game win/loss stats

Pillar 3: Engagement & Content

- Polls with customizable options, expiration, one-vote-per-user enforcement
- Like/Dislike reactions with instant toggle and count updates
- Nested threaded comments with up to 3 levels of indentation
- Built-in HTML5 media player: play/pause, volume, seek, fullscreen
- Custom video thumbnails (auto-generated or user-uploaded)
- Synchronized watch parties: host controls playback for all participants, late-joiner auto-sync, emoji reactions & chat sidebar

Results & Testing

40+

User Stories
Completed

48

E2E Tests
Passed

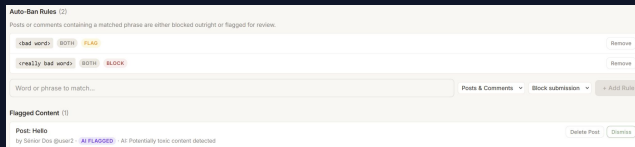
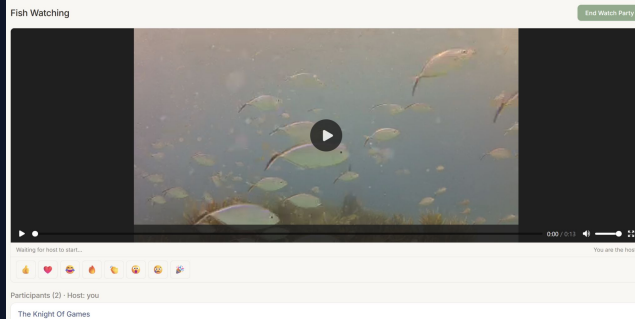
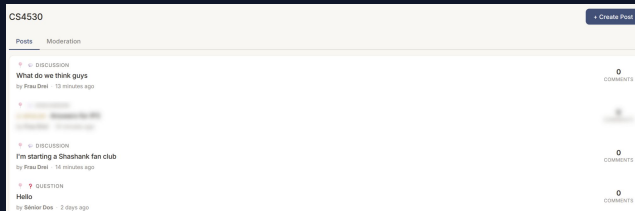
712

Unit Tests
Passed

90%+

Code
Coverage

Frontend: Playwright (Chromium) • Backend: Vitest • CI coverage reports



What's Next?

- Administrators can set up a tiered moderation system, with the page owner at the top and fully customizable permission levels below.
- A Preview Mode lets administrators see how a new banner will appear on both desktop and mobile before publishing.
- Your friends list shows which game pages each friend has joined, making it easy to spot shared interests.
- You can view a player's stats and rankings from external games through connected third-party platforms.

Agile Process & Sprints

- Sprint 0 (Feb 14–Feb 27)**
Repository setup, video hosting research, initial data model design. Completed as planned.
- Sprint 1 (Feb 28–Mar 13)**
Core post creation (frontend + backend), friend/DM system started. Shortened by spring break.
- Sprint 2 (Mar 14–Mar 27)**
Group chat, private game rooms, game page layout delivered. Media streaming infrastructure took longer than estimated.
- Sprint 3 (Mar 28–Apr 10)**
Carried extra scope: polls, likes, comments, media player, watch parties, friend features, AI moderation, player profiles all completed.

Technology Stack

Frontend	React, TypeScript, Vite, Socket.IO Client
Backend	Node.js, Express, Socket.IO, Zod Validation
Database	MongoDB (optional), Keyv in-memory store
AI/ML	Naive Bayes toxic content classifier (in-process)
Testing	Playwright (E2E), Vitest (unit), 90%+ coverage
Deployment	Render (deployed), monorepo with shared types

Key Takeaways

- Shared types package between client & server prevents type drift and hard-to-debug bugs
- Centralize auth in middleware instead of per-controller checkAuth() calls
- Hash passwords before production (bcrypt/argon2)
- Return error payloads from Socket.IO handlers for meaningful client feedback
- Distribute complex work (User Story 3) across sprints to avoid overloading the final sprint

Architecture

Three-layer architecture: Controller → Service → Repository



Key Design Decisions:

- Shared ChatRecord container across DMs, group chats, game rooms & watch parties
- Friend-gated social actions enforced at service layer
- Server-authoritative watch party sync with canonical state

CS4530 Final Project: “Nexus”

Team 207 Justin Liu, Brooklyn Cone, Daniel Pupke, Tyler Soong

Project Overview:

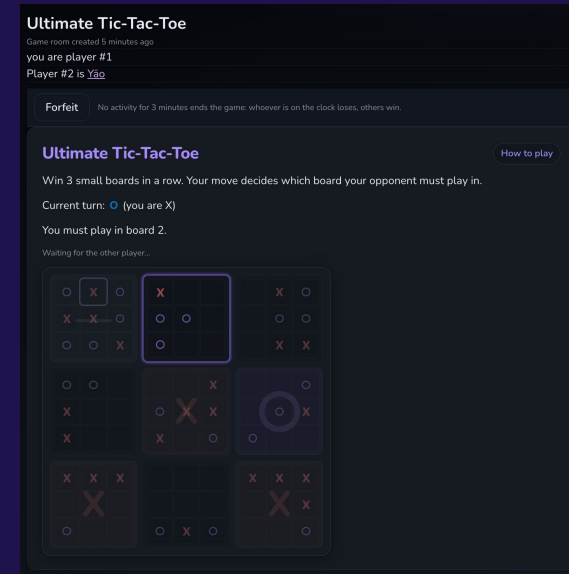
Nexus is a multiplayer browser-based gaming platform with comprehensive competitive infrastructure. Key features include a fully playable Ultimate Tic Tac Toe (UTTT) game, dynamic leaderboards with game-type filtering, persistent player profiles with statistics and achievement badges, match history with automated and step-through replay, and a complete social system featuring friends, game invites, rematch flows, and real-time notifications.

Live Demo & Repository

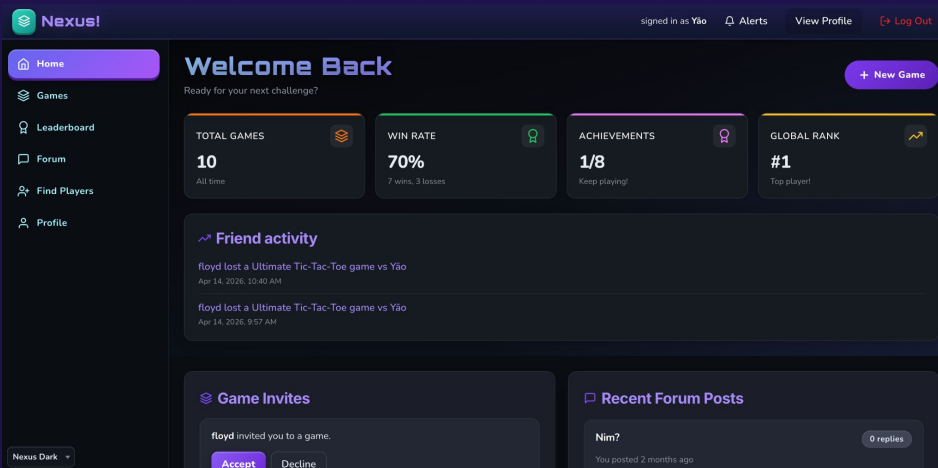
- **Demo:**
<https://spring26-project-spring26-project-group-qr74.onrender.com/>
- **GitHub Repo:**
<https://github.com/NEU-CS4530-S26/spring26-project-spring26-project-group-207>

What we Built:

We extended Nexus with Ultimate Tic Tac Toe using the existing game dispatch architecture so new games plug in consistently. We persist complete move histories so match history + replay can rebuild state move-by-move. Leaderboards use Mongo aggregation pipelines for dynamic, per-game-type rankings. Profiles combine match data into W/L/T, per-game breakdowns, win rate, and badge unlocks via an event-driven achievement path. Friends, invites, presence, and notifications run over Socket.IO. Rematch enforces two-sided confirmation before launching a follow-up game.



UTTT mid-game



Home Page

Stack:

React + TypeScript frontend; Express backend; MongoDB persistence; Socket.IO for realtime; deployed on Render.

Future Steps:

There are extensions we were not able to implement that would strengthen engagement, competition, and production reliability. With more time, we would deepen several systems beyond their current MVP scope. Examples include:

- UTTT AI opponent for offline practice and onboarding
- Data + query scaling
- Seasonal leaderboards + spectator / shared replays for community play



Team 208: Matthew Li, Manos Pepanidis, Ethan Man, Ronald Nguyen

PROJECT OVERVIEW

Multiplayer Gaming, Fully Social

NetPlay is a web-based multiplayer gaming platform designed to bring players together with a rich social infrastructure. Over Sprints 1-3, the team implemented a comprehensive suite of social features: user profiles, a friends system, direct messaging, game invitations, teams, and both individual and global leaderboards. Every feature was engineered for real-time responsiveness. Players see friend activity, game invites, and chat messages the moment they happen, no refresh required.

FEATURES

Profile Customization

**Profile for The Knight Of Games**
Total Points: 0
[Add Friend](#)

- Username: user0
- Account created 2 months ago
- Team: No Team

Bio
I am the greatest of all time

Friends

Friends
[My Friends](#) Requests Sent **1** Find Friends

Filter by name... A-Z

 **Sénior Dos** @user2 Last seen: 4 days ago [Invite to Game](#) [Remove](#)

Teams

Team: NetPlayers [Administrator](#)

[Roster](#) [Stats](#)

Team Stats

7 Wins

6 Losses

1.17 W/L Ratio

13 Games Played

0.54 Avg Score

Member Contributions

 Yao (@user1) 4W / 3L 7 games 4 pts

 Sénior Dos @user2 3W / 3L 6 games 3 pts

Leaderboards

Global Leaderboard (all-time)
Global All Time
[Individual](#) [Team](#)

RANK	USERNAME	TEAM	POINTS
1 ▲	Yao (@user1)	NetPlayers	4
2	Sénior Dos (@user2)	NetPlayers	3
3	Frau Drei (@user3)	No Team	1
4	test2 (@test2)	No Team	1

Direct Messages

Direct Messages

@friend's username... [Start](#)

 user2 I've invited you to a game!...

 user2 user2 invited you to a nim!
[JOIN GAME](#) [DECLINE](#)
Apr 9, 2:52 PM

 user1 How are you doing?
10:51 AM

 user2 I'm good, how are you?
10:13 AM

 user1 Great!!
10:51 AM

 user1 Let's play
10:51 AM

TECH STACK + DESIGN DECISIONS

We extended the existing GameNite codebase, a TypeScript monorepo consisting of a React + Vite frontend, an Express + Node.js backend, and a shared types package used by both. New REST endpoints handle friend requests, DM conversations, team management, and leaderboard aggregation, while Socket.IO broadcasts real-time events: friendRequest, dmNewMessage, dmNotification, and leaderboardUpdated, to connected clients. Notification state is shared across components using React Context, avoiding prop drilling between the always-mounted layout and individual page components. Our CI pipeline runs TypeScript checks, ESLint, 86 Vitest unit tests, and 61 Playwright end-to-end tests on every pull request, with the application deployed to Render.com.

FOR THE FUTURE

Looking ahead, the most exciting opportunity is expanding the game library. With the full social infrastructure in place (friends, teams, leaderboards, and real-time invites), adding games with more strategic depth, such as a card game, chess, or backgammon, would give the competitive features far more meaning and better showcase the platform's potential. We would also prioritize implementing persistent authentication, as the current in-memory session is lost on page refresh, which is the most impactful usability improvement we could make. Also, a friend activity feed showing when friends finish a game, join a team, or climb the leaderboard would deepen social engagement and give users a reason to keep coming back to the platform. Lastly, we would love to look into implementing a mobile-friendly (iOS/Android) version of NetPlay in the future.



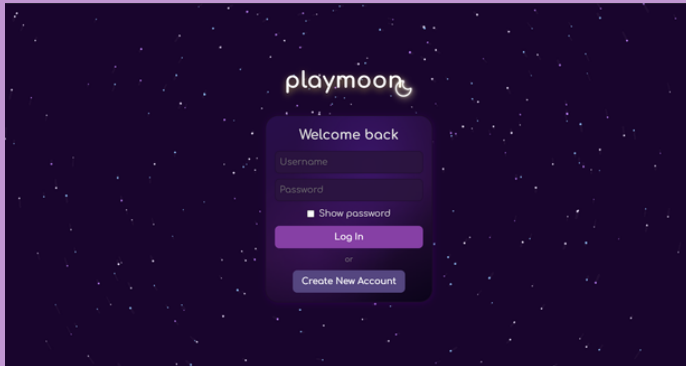
PlayMoon

Noelis Aponte, Daniel Fernando-Sorondo, Kaelynn Lackey, Maciej Blaszcak

How to Play

Click this link: <https://spring26-project-group209.onrender.com>

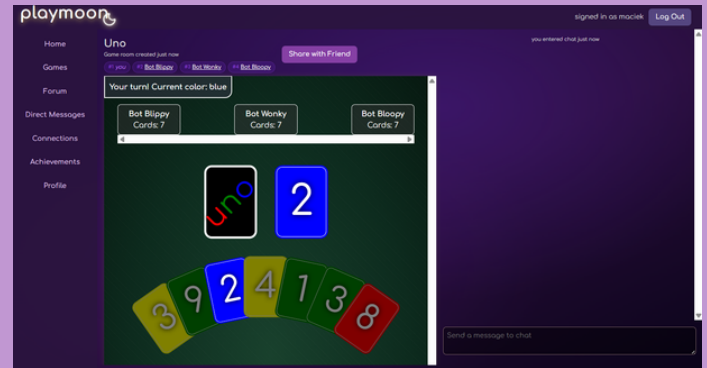
Wait until you see this screen & sign up to play!



Uno



Play a relaxing game of UNO with some friends or play against some bots to hone your fantastical UNO skills!

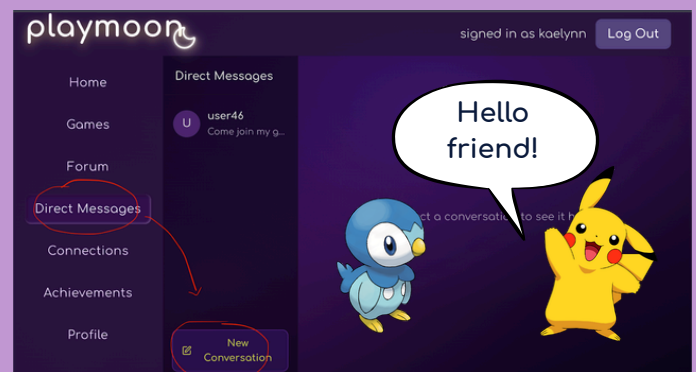
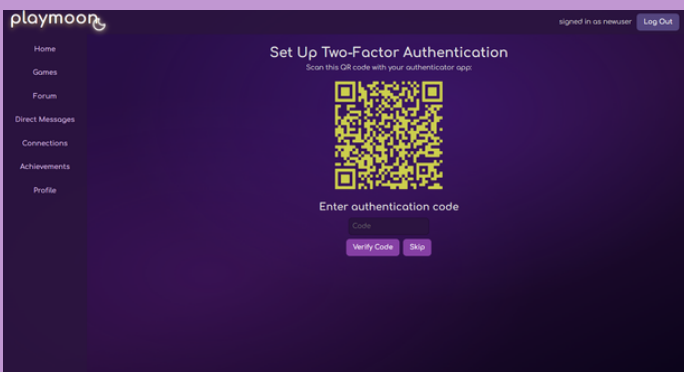


Security and Authentication

Have a secure account through two factor authentication and secure passwords. Enjoy private games with friends!

Social Features

Follow and friend people to create a community. DM them and invite them to games!



Our Inspiration

The original GameNite offered a few features aimed at quality and engagement, but it lacked fun, relevant games, community-building tools, and social features, resulting in a site that felt both ugly and boring. To reimagine it, we drew inspiration from Illumise, the firefly Pokémon known for its playful nature and its ability to create beautiful light patterns alongside Volbeat. Because Illumise forms genuine connections, whether with other Pokémon or its trainer, we wanted to bring that same spirit to our platform, giving users what we call the "Illumise Experience." This vision led us to create PlayMoon, building on the GameNite foundation by adding UNO, social features, and robust security and authorization to foster a more connected, enjoyable experience.

Future Work

- Add more games for a more robust user experience.
- Improve the capabilities of direct messaging to include images.
- Add a CAPTCHA for greater security.





CS4530 Final Project: “Playbook”

Group 210: Aditya, Reid, Mak, Ethan

Our Features

PlayBook enhances community engagement through a robust reputation and discovery system. Users earn and lose karma through upvotes and downvotes on their posts and comments, with their karma history visualized as an interactive graph on their profile. A global search ties it all together, letting users discover posts, games, and other users through relevance-scored results. The platform also prioritizes communication and creativity. The Friends system lets users build their social network through a full friend request workflow for updates. Group Chats allow any number of users to communicate in real time. Finally, an AI Assist feature powered by Groq provides real-time writing suggestions while drafting forum posts, making it easier to articulate ideas and contribute to discussions.

Create new post

TITLE

Why Game of thrones is the best show?

TAGS (UP TO 3)

strategyopeningsendgamesanalysisgeneral

POST CONTENTS

Game of thrones is the best show because of its intricate characters and unpredictable plot twists that keep you hooked from start to finish, and the way it explores the complexities of human nature is just amazing.

POST

What’s Next?

For future work we want to add more multiplayer games and enhance security by adding OAuth.

↑
|
↓

weather

by You 3 Apr 2026

the weather looks great outside

You 6d ↑ | ↓

adi

Yao 6d ↑ | ↓

hi

Our Tech Stack & Design

PlayBook is built on a TypeScript monorepo with three workspaces: a React frontend, Express backend, and a shared package for common types and validation schemas enforced with Zod. The client is bundled with Vite and uses React Router for navigation, Axios for HTTP requests, and Socket.IO for real-time updates across features like group chat messaging, friend notifications, and live game state. The server runs on Node.js with Express and persists data in MongoDB, with Keyv used for caching. New features — karma, leaderboards, search, friends, group chats, and AI assist — were layered onto the existing backend via dedicated service and controller modules, and surfaced in the frontend through custom React hooks and page components. The AI Assist feature integrates with the Groq API via the OpenAI-compatible SDK to provide real-time writing suggestions while drafting forum posts. Our continuous integration pipeline runs an automated test suite using Vitest for unit and integration tests and Playwright for end-to-end tests, with ESLint and Prettier enforced across all workspaces.

Leaderboard

Search user

Sort/Filter

RANK	PLAYER	WINS	LOSSES	TIES	PLAYED	WIN RATE	HRS	KARMA
1	You	1	1	0	2	50.0%	0h	3
2	Sénior Dos	1	2	0	3	33.3%	0h	0
3	Frau Drei	1	0	0	1	100.0%	0h	0
4	The Knight Of Games	0	0	0	0	0.0%	0h	0
5	Yāo	0	0	0	0	0.0%	0h	1

Our demo is available [here](#). Our code is available [here](#)

NightOwl

Real-time multiplayer · Party games · Living player profiles

Jack Yu
Tanmay Demble
Benjamin Lam
Ryan Amiri

ABOUT

NightOwl is a real-time multiplayer platform that transforms how people play party games online. Built with a scalable TypeScript stack, it ships Codenames with AI support, full match replay, and adaptive player profiles that get richer with every session.

Codenames: spymaster view

Word Pack: movies
Team Assignment: Random
AI Guesser: Enabled
You are Team 1 **spymaster** (user1111).

monster	cyborg	theater	titanic	matrix
shadow	sword	flash	pilot	camera
legend	empire	horror	jungle	royal
beast	laser	clown	cinema	animation
prince	epic	action	remake	hero

Codenames

Team-based party game where spymasters give one-word clues to guide teammates across a 5x5 word grid. Supports AI guessers and custom word packs.

Game Replay

Every game is recorded as turn-by-turn snapshots. Scrub through a timeline, jump to any move, and review the full match in read-only mode.

Dynamic Identity

Profiles auto-generate behavior traits from gameplay stats: win rate, pace, sociability, and game preference. Presence state updates live via WebSocket.

Replay: step through any past game

Game Id: 62df3527-5a8d-490f-9268-ffdc1cce75c

Players: ● you, ● user2222

Turn 7 of 8

Timeline

Jump to turn

7

Jump

By: user2222

Action: placed a mark on square 8

Play

Previous Turn

Next Turn

Turn 3: placed a mark on square 5

Turn 4: placed a mark on square 9

Turn 5: placed a mark on square 3

It is currently user1111's turn (X).

X		O
	O	
X	O	X

Watching replay mode (read-only).

Player profile: dynamic identity

Dynamic identity

COMPETITIVENESS

developing

Still building momentum in rated games.

SOCIABILITY

unknown

No game-chat activity yet.

PACE

quick

Usually finishes games quickly.

PREFERENCE AFFINITY

explorer

Spreads play time across different games.

Tech Stack

- TypeScript monorepo
- React 19 + Vite + React Router
- Express 5 + Socket.io
- MongoDB via Keyv
- Vitest + Playwright E2E

Architecture

- Shared DTOs across all layers
- REST for CRUD, Socket.io for events
- Immutable server-side game state
- Replays stored as turn snapshots
- Dependency injection throughout

What's Next

- More games: Pictionary, Trivia
- Leaderboards and tournaments
- Spectator mode
- AI opponents for async play
- Plugin architecture for games

Team 212 - Pixel Cafe

Amine Nazih, Joseph Trisnandi, Allie Clark, Akshitha Bhashetty

Overview

GameNite is a competitive browser-based game platform where players compete in real-time games, climb a global leaderboard, earn chips for winning, and spend them on cosmetic upgrades. Players can view game tutorials, join bracket-style tournaments, follow friends' rankings on a friends-only leaderboard, and customize their profiles with frames, badges, and titles.

First time playing Nim?

Want a quick rules refresher before you continue to the game?

Nim



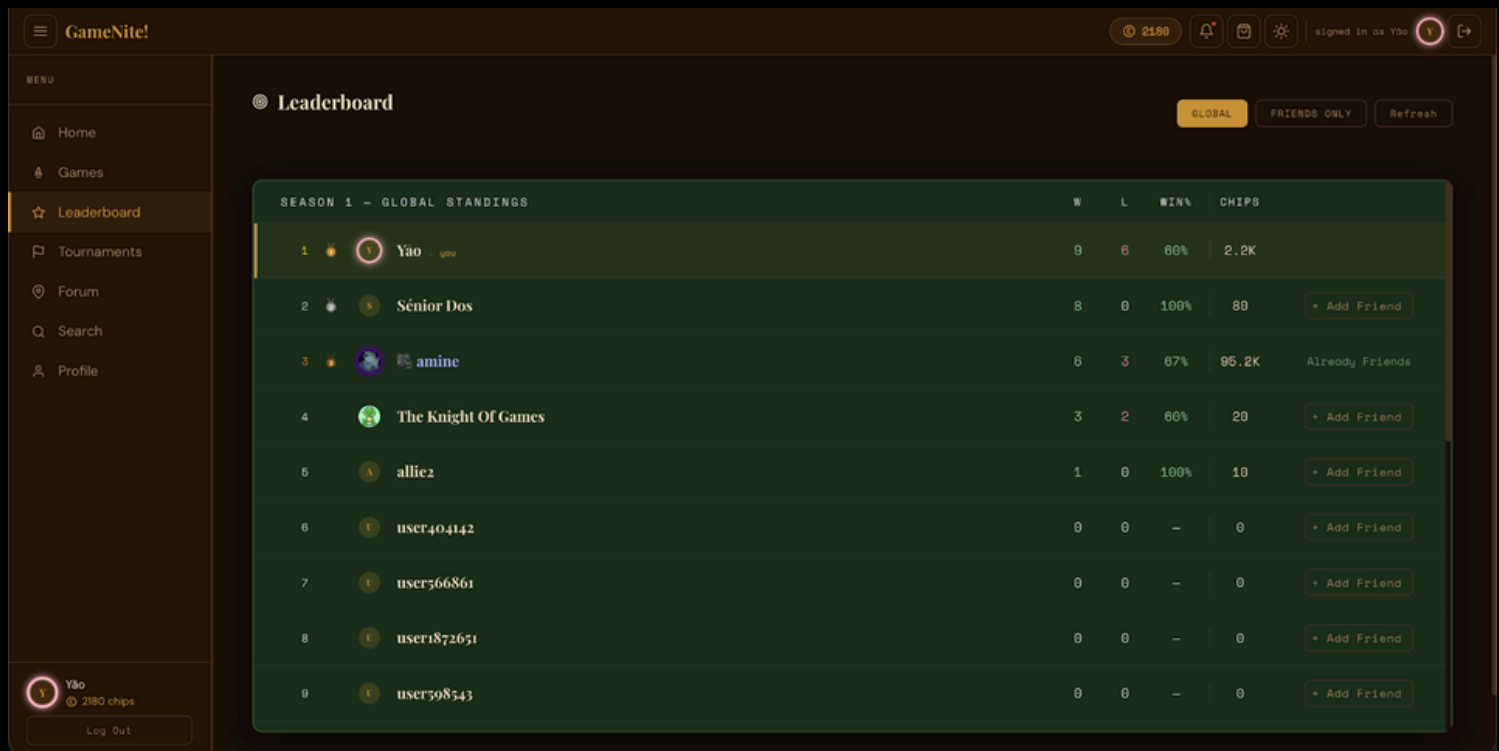
Take 1-3 each turn. Avoid the last one.

View Rules

Skip

Website Link

Source Repo Link

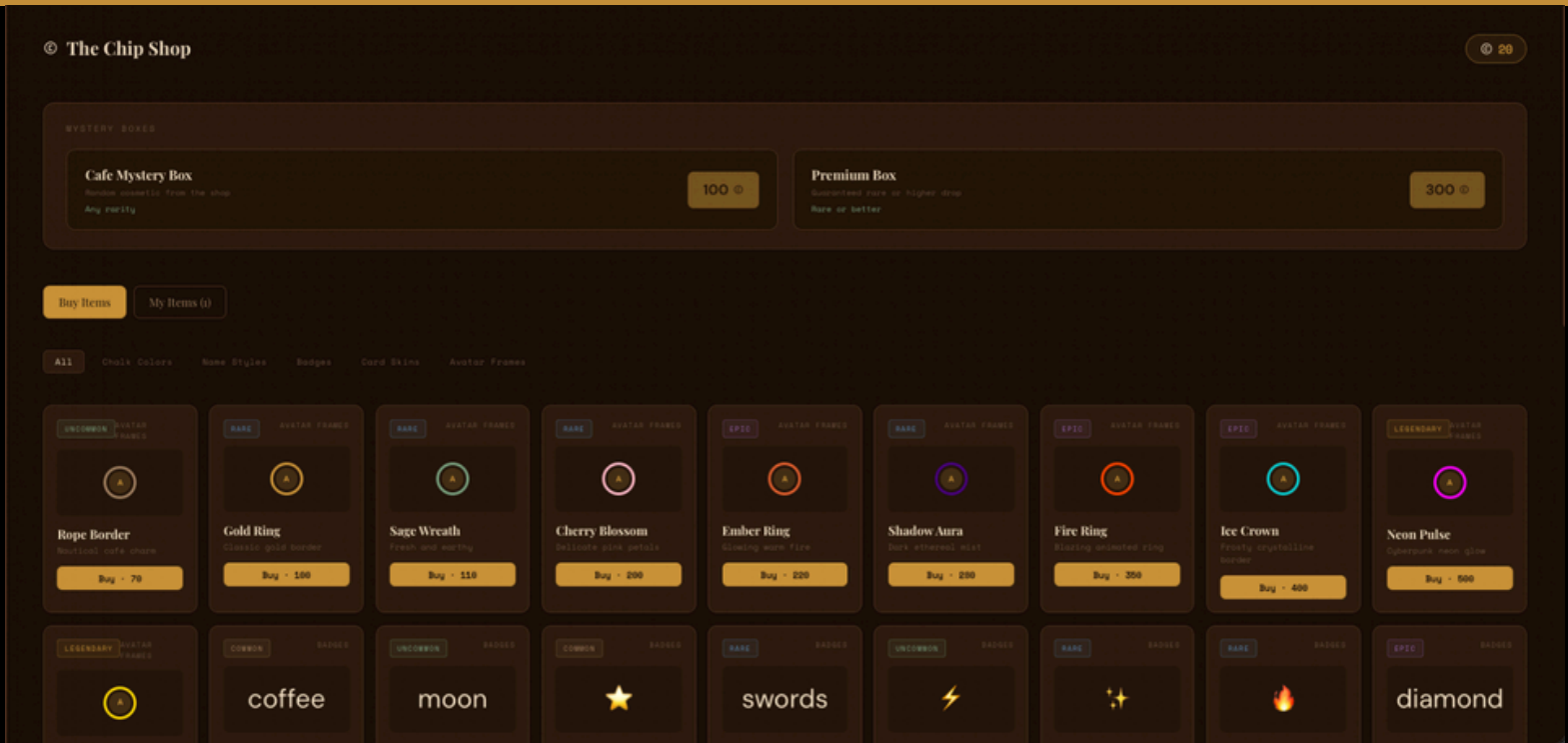


What's Next

- Per-game leaderboards - separate rankings for each game type
- Mutual friends display - show how many friends you share with another player on their profile
- Gifting system - send cosmetic items to friends
- Refactor - extract the leaderboard rank computation into a background job rather than a synchronous RPC call, to improve game completion latency at scale

Design Decisions

- Frontend: React + TypeScript, React Router, Vite
- Backend: Node.js + Express, Socket.io for real-time game events
- Database: Supabase (PostgreSQL) with Row Level Security; player ranks are recomputed via a Postgres RPC after each game outcome so rankings stay consistent without polling
- Auth: Username/password via bcrypt + OAuth (Google) via Supabase Auth
- Testing: Vitest + React Testing Library for unit tests; Playwright for end-to-end flows
- CI/CD: GitHub Actions on every PR

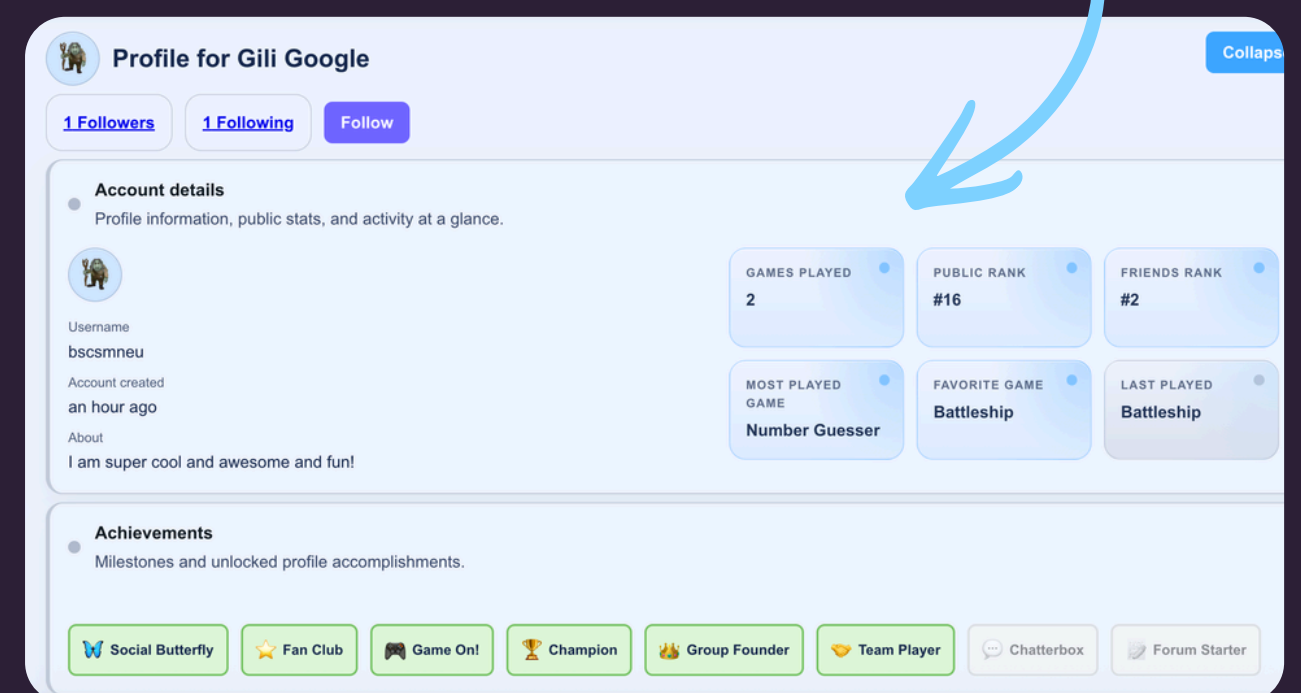




Dana Abdo, Gili Cohen,
Julien Mohatarien, Ethan Szeto

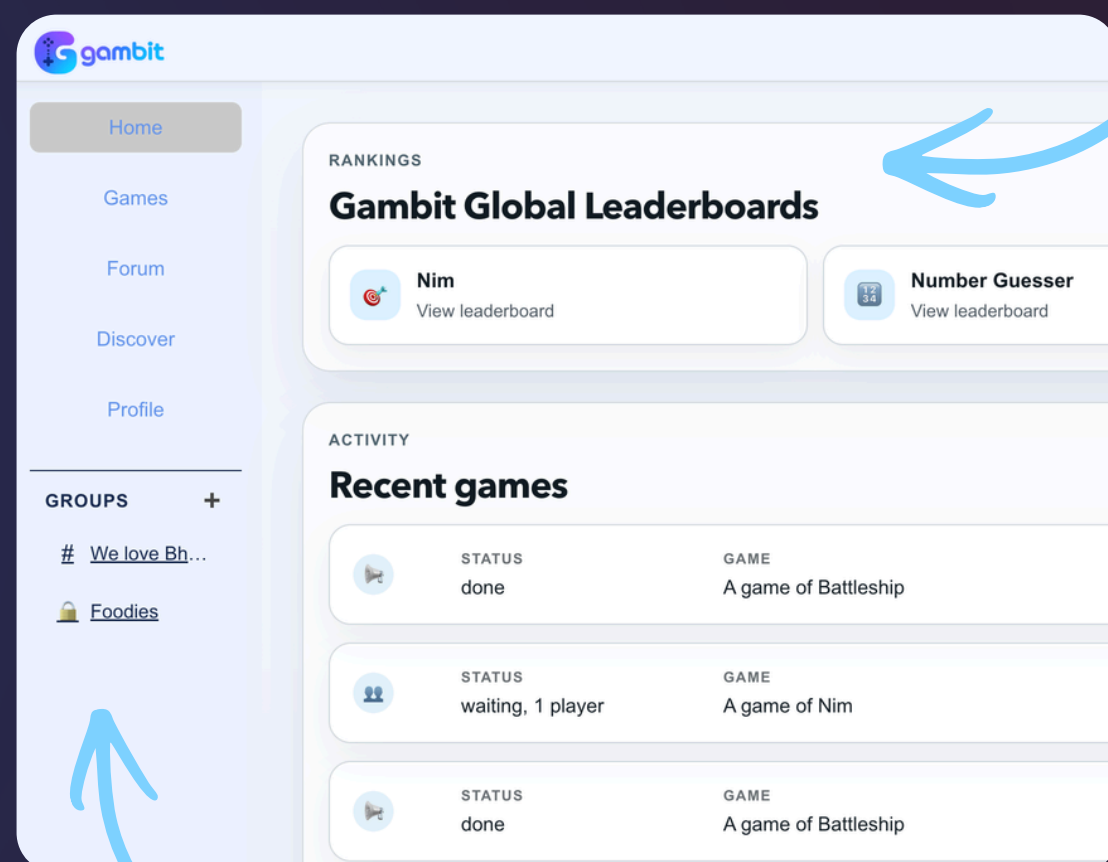
Gambit is a **social gaming platform** designed to turn one-off matches into lasting connections and communities. It enables players to build relationships through **friends and followers**, organize gameplay within public or private groups, and engage in socially contextual experiences that persist over time. With features like messaging, shared game activity, competitive leaderboards, and both multiplayer and bot-based gameplay, Gambit supports continuous engagement, skill development, and meaningful interaction—transforming gaming from isolated sessions into a connected, community-driven experience.

User Stats. Profile Views, and Avatars



Collectable User Achievements

Global Game Leaderboards



Public/Private Groups



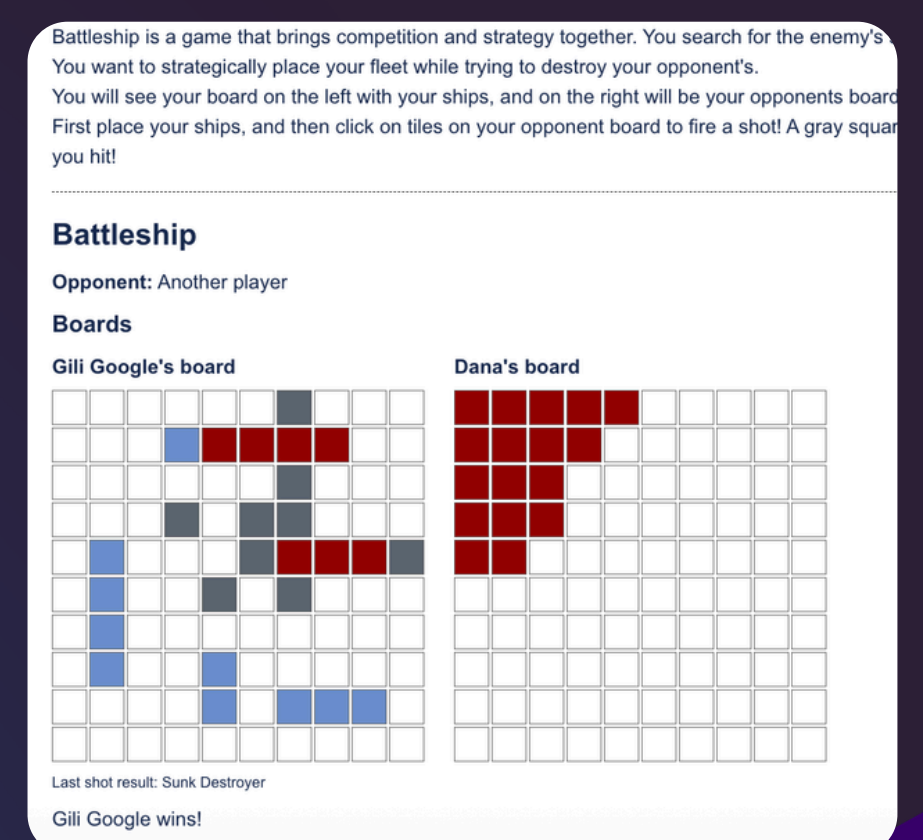
Deployed with CI/CD on Railway and MongoDB Atlas using a monorepo project architecture.



Utilizes a React frontend with an express backend. The backend follows the CSR pattern. Google SSO for authentication and session persistence

Looking ahead, we'd push the platform further by adding Microsoft/Northeastern SSO for seamless campus login, advanced per-user stats and trend graphs, Battleship power-ups for deeper gameplay strategy, and a full multi-group tournament bracket system.

Battleship Game + AI Opponent



<https://spring26-project-group-213-production.up.railway.app/login>



<https://github.com/neu-cs4530/spring26-project-group-213>

Arcadia!

Created By Group 214:
Aidan Essig, Aparajitha Karipineni, Nicholas
Yang, Timothy Qian

What is Arcadia?

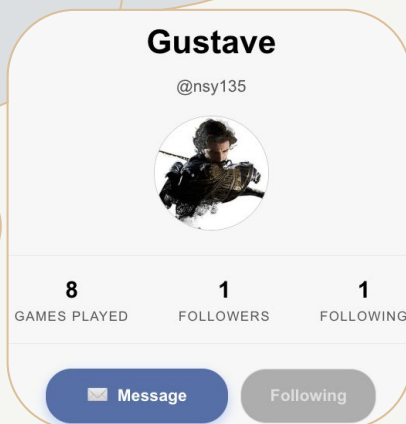
Arcadia is an engaging,
social and accessible
web-based gaming platform.

Arcadia's Goals

- Rich User Interaction
- Game Variety
- Seamless Mobile Play

Tech Stack

- Full-stack Typescript application
 - React: frontend
 - Node.js + Express: backend
- Socket.io: real-time interactions (gameplay, chat, etc.)
- MongoDB: data storage
- Session management and Google OAuth: authentication
- Render; Deployment
- GitHub Actions: CI
- GameNite: Source Codebase

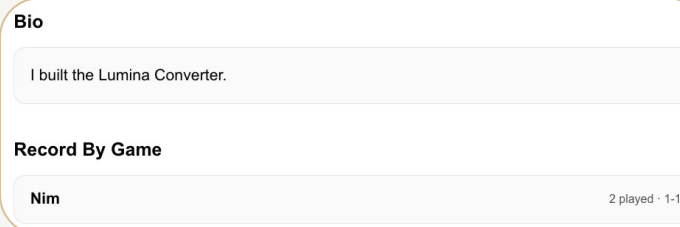
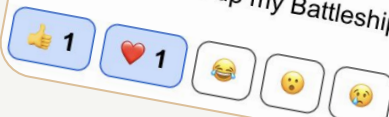


Features!

- Profile Personalization!
- Game record displays!
- Follow/message system!
- React to other users' posts!
- New games: Wordle and Battleship!
- Accessibility and fully responsiveness on mobile devices!
- Dark Mode!

Anyone around this evening to play?

Looking to brush up my Battleship skills!



What's Next?

- New games: increase engagement and variety
- Notifications + other social features: user interaction
- UI/UX refinements: user satisfaction
- Toggle game records to be public or private: additional user satisfaction
- See display names of followers: foster larger community



Design Decisions

- Dark Mode: simple toggle with global theme context for instance changes and consistency between sessions
- Account creation/login via SSO to minimize credential management overhead
- Private communication: profile-driven system
- Curated 5-emoji set for forum reactions with uniqueness constraint to prevent spam
- Battleship with multiple difficulty levels that allowed players to place their own ships and attack opponents' ships
- Wordle with multiple difficulty levels that changed secret word lengths and number of guesses
- Split user information into summary and profile types to maintain SRP
- Store game records as fields, followers in arrays

Project Repo: [Click Here!](#)

Hosted App: [Click Here!](#)

ABOUT THE PROJECT

Playrift is a platform for turn-based games, forums, and group competition. Players can challenge opponents to Nim or Number Guesser, track progress through leaderboards and game history, and form communities through groups, with moderation tools for group leaders and interaction controls (block, mute, report, hide) for every user.

LINKS

[▶ Demo Site](#)
g215-playrift.onrender.com

[Source Repository](#)

TECH STACK & DESIGN DECISIONS

Stack

Frontend: React + Vite client

Backend: Node.js + Express server

Database: MongoDB

CI: Github Actions

CD: Deployed on Render

API & Data Model

New /groups and /dm resources with full CRUD endpoints

New Chat, Message, DM, Group, and LeaderboardConfig models

Shared messaging powers DMs, group chat, in-game chat

Architecture

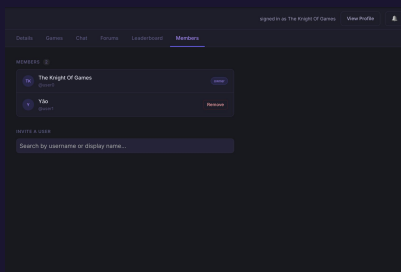
Extended User, Thread, and Comment models for moderation and privacy
Games and Threads optionally scoped to a Group

Agile workflow: 4 sprints, GitHub Projects

KEY FEATURES

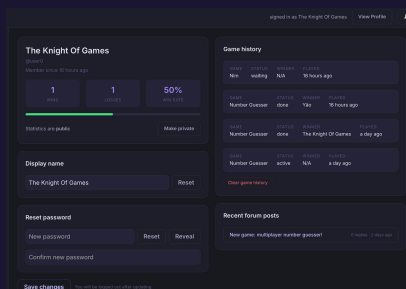
1 Groups

Create groups with chat, forums, exclusive games, and a leaderboard. Leaders moderate members, messages, and forum posts; users can be invited, removed, or readmitted.



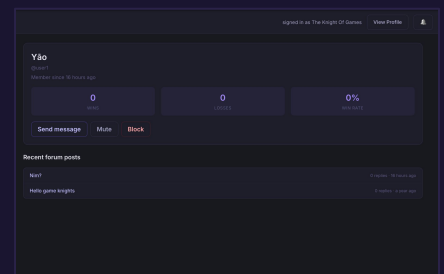
2 Gameplay Stats

Profile tracks wins, losses, win rate, and win streaks with a chronological game history. Stats can be made private, hiding them from others and limiting your own leaderboard access.



3 Moderation

Block, mute, or DM any user, with unread-message notifications. Report group forum posts, hide your own posts, clear chat history, and add emoji reactions to any message.



WHAT'S NEXT?

Recurring group events

Scheduled tournaments and weekly game nights sent to a group, with automatic bracket generation.

In-game badge ranks

Visible rank indicators next to usernames based on total wins, displaying player skill levels at a glance.

Friends system

Send and accept friend requests, with a dedicated friends list and quick-challenge shortcuts.

Refactor: unified notifications

Consolidate DM, report, and group notifications behind one pub-sub service to simplify the frontend.